STUDENTS' TECHNICAL COUNCIL

sutra management consultancies

14-16 APRIL 2023

TECHSHILA

INTER BHAWAN TECH MEET

Consulting & Analytics

#6

Sutra Management Consultancies' Case Challenge

A bank's compliance department has integrated an Anti-Money Laundering(AML) solution to detect potentially suspicious activities associated with money laundering. It works on specific scenarios like large cash transactions, transactions from multiple locations & suspicious countries, etc. However, the bank is facing high false positives, leading to an increased investigation load. The client has approached us to help them reduce the false positives generated by the AML solution.

Problem Statement

Anti Money Laundering

AML stands for Anti-Money Laundering, which refers to a set of laws, regulations, and procedures that financial institutions and other regulated entities must follow to prevent, detect, and report money laundering activities. Money laundering is the process of concealing the origins of illegally obtained money, typically by transferring it through a series of transactions and bank accounts to make it appear as if it was obtained from a legitimate source.

A bank is required to monitor transactions using specific rules to detect any potentially suspicious activities related to money laundering. These scenarios are designed to identify certain types of risks that may help banks to identify and flag suspicious transactions to detect patterns of illegal activities and report such activities to appropriate authorities.

AML Scenarios

AML scenarios are specific types of financial transactions or activities that are commonly associated with money laundering or other illegal activities. Here are some examples of AML scenarios:

1. Large cash transactions – Transactions involving large amounts of cash, especially by entities with no prior history of transactions, are often considered suspicious and subject to AML scrutiny.
2. High-risk countries - Transactions with individuals or entities based in countries known for money laundering or terrorist financing activities may be flagged for AML review.

Details of some scenarios generating a high number of alerts for the bank are mentioned in the Appendix.

Problem Statement

False Positives in AML
The AML solution generates daily alerts based on customers' transaction behaviour to detect potential money laundering. However, investigations often reveal that these alerts are for customers not involved in money laundering activities, which poses a challenge for the banks as it results in an increased workload for their investigation teams.

Productivity in the AML Context
Productivity can be defined as the ratio of productive alerts to total alerts generated by the AML solution. Productive alerts are those that result in a confirmed case of money laundering or other illegal activity, while non-productive alerts are those that are found to be false positives or do not require further investigation.
Example:
AML solution generates 100 alerts in a day and only 10 are found to be productive, the productivity ratio would be 10/100 or 10%. By optimizing the AML solution to generate fewer false positives and more productive alerts, financial institutions can increase their productivity and reduce the workload on their investigation teams while effectively identifying and preventing money laundering activities.

Tasks

1. **Propose an analytical approach:** Provide a detailed analytical approach to fine-tune the current AML solution and optimise productivity

Tasks

2. **Recommend best possible approach:** Analyse scenarios that are generating high false positives, optimize productivity and recommend the best possible approach with a champion challenger situation
3. **Propose an implementation strategy:** Propose a detailed implementation strategy for the recommended solution

Evaluation

There are some key points on which the Problem statement will be judged -

1. **Effectiveness** of the proposed solution in reducing the number of false positives **(50%)**
2. **Feasibility** of the solution within bank's existing systems and processes **(30%)**
3. **Scalability** of the solution to accommodate future growth in the bank's customer base and transaction volume **(20%)**

Participation

Participation details related to the challenge:

1. The maximum team size is 5 members.
2. Contact your bhawan's contingent leader to participate.

Submission

You need to submit a detailed presentation (.pdf format) summarizing all the relevant deliverables

Deadline: 14th April , 11:59 PM

Appendix

The details of a few scenarios which are generating the highest false positives for the bank is attached below for better understanding. The scenarios work on specific parameters, such as checking a customer's last five days' transactions and comparing them with a threshold value.

Scenario 1 : Large cash deposit transactions

Category	Cash Activity
Severity	Scenario
Frequency	Daily
Replication (Business Days)	1 Day
Subject	Customer
Short Description	Large Cash Deposits
Long Description	A customer’s daily total of cash deposits exceeds a threshold.
Business Note	This scenario calculates the sum of a customer’s cash deposits. If the amount exceeds the threshold, a match is found. This scenario examines only cash deposits for a single day.
Days Considered	Not Applicable

Appendix

Scenario 2 : Multiple Location Usage

Multiple locations refer to the customer making transactions at bank's different ATMs and branches

Category	Structuring and Obfuscation
Severity	Scenario
Frequency	Daily
Replication (Business Days)	1 Day
Subject	Account
Short Description	Multiple Location Usage
Long Description	An account has multiple transactions at multiple locations on a single day
Business Note	This scenario examines account transactions for activity at multiple locations. If an account's number of locations and total transaction amount exceed the threshold, a match is found.
Days Considered	Not Applicable

Scenario 3 : High Risk Countries

There are certain countries with weak money laundering laws. Bank maintains a list of all such countries and tag them as high risk-countries

Appendix

Category	Watch List entities
Severity	Scenario
Frequency	Daily
Replication (Business Days)	5 Day
Subject	Customer
Short Description	High-Risk Countries
Long Description	A customer engages in transactions to or from high-risk countries. Administrators at your site maintain a list of high-risk country codes.
Business Note	This scenario examines transactions involving countries by checking a list of high-risk countries codes. If a transaction is found, then that customer;s total amount of high-risk country transactions is calculated for the time period. If the total amount exceeds the threshold, a match is found. This scenario considers business days
Days Considered	Business

References

1. Introduction to AML
2. https://www.investopedia.com/terms/a/aml.asp
3. Champion-Challenger Technique